



LIMITING REAGENTS 1

In each example one of the reactants is in excess. Work out how many moles of the products are formed in each case.

<u>1</u>	CaO	+	H₂O	→	Ca(OH)₂	
a)	2 mol		3 mol		2 mol	
b)	10 mol		8 mol		8 mol	
c)	0.40 mol		0.50 mol		0.40 mol	

<u>2</u>	2Ca	+	O₂	→	2CaO	
a)	2 mol		2 mol		2 mol	
b)	10 mol		2 mol		4 mol	
c)	0.50 mol		0.20 mol		0.4 mol	

<u>3</u>	2Fe	+	3Cl₂	→	2FeCl₃	
a)	3 mol		3 mol		2 mol	
b)	12 mol		15 mol		10 mol	
c)	20 mol		40 mol		20 mol	

<u>4</u>	TiCl₄	+	4Na	→	Ti	+	4NaCl
a)	4 mol		4 mol		1 mol		4 mol
b)	2 mol		10 mol		2 mol		8 mol
c)	0.5 mol		1 mol		0.25 mol		1 mol

<u>5</u>	C₂H₅OH	+	3O₂	→	2CO₂	+	3H₂O
a)	15 mol		30 mol		20 mol		30 mol
b)	0.25 mol		1 mol		0.5 mol		0.75 mol
c)	3 mol		6 mol		4 mol		6 mol

<u>6</u>	N₂	+	3H₂	→	2NH₃	
a)	3 mol		6 mol		4 mol	
b)	0.5 mol		0.9 mol		0.6 mol	
c)	6 mol		20 mol		12 mol	

<u>7</u>	4K	+	O₂	→	2K₂O	
a)	10 mol		2 mol		4 mol	
b)	6 mol		4 mol		3 mol	
c)	0.50 mol		0.20 mol		0.25 mol	